



Gnanamani College of Technology

(AUTONOMOUS)

Approved by AICTE, New Delhi & Affiliated to Anna University, Chennai
NH-7, A.K. Samuthiram, Pachal – 637018, Namakkal (Dt.), Tamilnadu, India.
Web : www.gct.org.in, E-Mail : info@gct.org.in, Contact No. 7598293888, 7598293999

Institute Vision:

To emerge as a globally recognized technical institution to produce ethical engineers, researchers, administrators and entrepreneurs

Institute Mission:

- To provide state-of-the-art infrastructure to create an effective learning environment for students
 - To collaborate with leading industries and academia to empower students to meet global standards
 - To foster an enterprising environment that encourages innovation and entrepreneurial activities among students
-

Department Vision: (IT)

To be the department that imparts professional computing training and makes competent engineer to work in the emerging areas of information technology field.

Mission Statements: (IT)

- To prepare competent engineers and adapt to the dynamic needs of industries.
 - To pave way to the enrichment of knowledge and skills using latest technologies in the diverse domain of information technology.
 - Inculcate strong ethical values and professionalism to serve society while concurrently updating knowledge and skills through life-long learning.
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PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

Graduates of Information Technology will

- **PEO-1:** Be proficient in utilizing the fundamental knowledge of various streams in engineering and technology.
- **PEO-2:** Think logically and to pursue lifelong learning to understand technical issues related to computing systems and to provide optimal solutions.
- **PEO-3:** Design and develop hardware and software system by understanding the importance of social, business and environmental needs in the social contexts.

DEPARTMENT OF INFORMATION TECHNOLOGY

PROGRAM OUTCOMES (POs)

- 1.Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
- 2.Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- 3.Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- 4.Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

Graduates of the program will be able to

- **PSO-1:** Apply the mathematical and the computing knowledge to identify and provide solutions for computing problems.
- **PSO-2:** Design and develop computer programs in the areas related to algorithms, networking, web design and data analytics of varying complexity.

Ex.No:1	DATA EXPLORATION AND INTEGRATION WITH WEKA
Date :	

AIM :

To perform Data exploration and integration using WEKA tool in Data warehousing .

PROCEDURE :

a). Data exploration using WEKA tool

- Open start → programs → Accessories → Notepad++
- Type the following sample dataset program on Notepad++ for creating Weather table.
- After the weather table program created save the file name in .arff(attribute-relation file format) file formate .
- For Data exploration Open WEKA tool the dialog box displayed on the screen
- Click Explorer → preprocess
- In Preprocess it shows many options. select " open file " option and open the file with .arff file formate.
- The attributes of our program displayed on the screen with current relation , visualize all data's.
- To view the table go to Edit option the viewer shows the table with attributes and datas.

PROGRAM :

For weather dataset :

@relation weather

@attribute outlook {sunny,overcast,rainy}

@attribute temperature numeric

@attribute humidity numeric

@attribute windy {true,false}

@attribute play {yes,no}

@data

sunny,85,85,false,no

sunny,80,90,true,no

overcast,83,86,false,yes

rainy,70,96,false,yes

rainy,68,80,false,yes

rainy,65,70,true,no

overcast,64,65,true,yes

sunny,72,95,false,no

sunny,69,70,false,yes

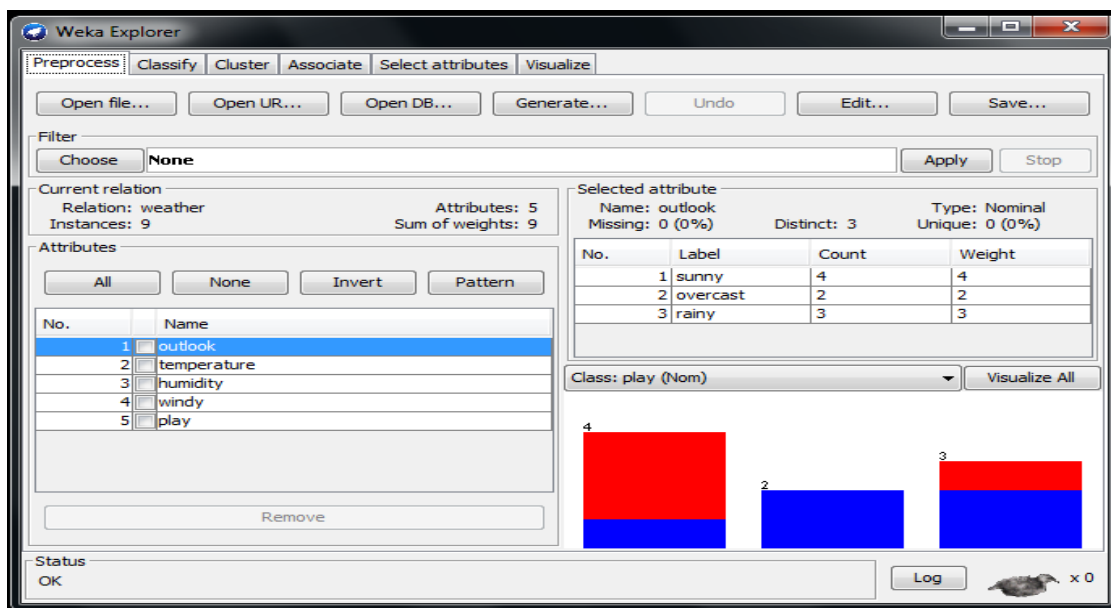
overcast,75,90,true,no

OUTPUT :

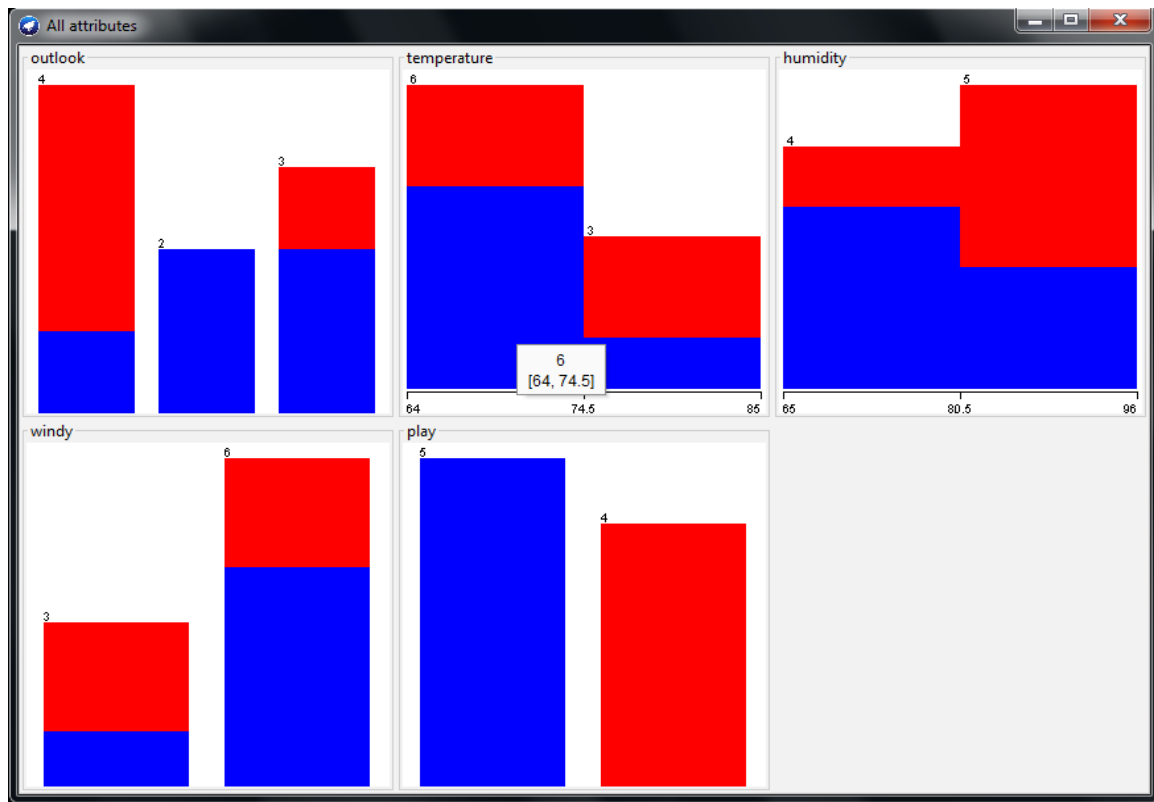
WEKA TOOL :



DATA EXPLORATION:



VISUALIZATION :



DATA VIEWER :

Viewer

Relation: weather

No.	1: outlook Nominal	2: temperature Numeric	3: humidity Numeric	4: windy Nominal	5: play Nominal
1	sunny	85.0	85.0	false	no
2	sunny	80.0	90.0	true	no
3	overcast	83.0	86.0	false	yes
4	rainy	70.0	96.0	false	yes
5	rainy	68.0	80.0	false	yes
6	rainy	65.0	70.0	true	no
7	overcast	64.0	65.0	true	yes
8	sunny	72.0	95.0	false	no
9	sunny	69.0	70.0	false	yes

Add instance Undo OK Cancel

b) Data integration using WEKA tool

PROCEDURE :

- Open start → programs → Accessories → Notepad++
- Create two dataset program on Notepad++ for creating Weather tables (create two weather tables as weather & weather1 & In the weather table it should have same attributes in both weather tables) and saved in same folder and also same location(D: or E: drive).
- Also create empty dataset with out any program in the same folder with .arff file formate.
- After the weather tables and empty file created save the file name in .arff (attribute-relation file format) file format.
- For Data integration Open WEKA tool the dialog box displayed on the screen.
- Click Simple CLI → Enter commands in the textfield at the bottom of the window.
- In commend field by using the following command it combine the two datasets merged in to single dataset.
- The commend was :

java weka.core.Instances append 1st weather table file location.arff 2nd weather table file location.arff> result file location.arff

- For example:

```
java weka.core.Instances append D:/weather.arff D:/weather1.arff > D:/result.arff
```
- After type the command click Enter button.
- Then repeat the data exploration process and open the file name result.arff (we created empty file)
- After we integrating the two data sets it merge in to single data set and show the result in the result.arff file.
- To view the table go to Edit option the viewer shows the table with attributes and datas of two datasets.

PROGRAM :

For weather1 dataset :

@relation weather1

@attribute outlook {sunny,overcast,rainy}

@attribute temperature numeric

@attribute humidity numeric

@attribute windy {true,false}

@attribute play {yes,no}

@data

rainy,68,80,false,yes

rainy,65,70,true,no

overcast,64,65,true,yes

sunny,72,95,false,no

sunny,78,68,false,yes

overcast,68,87,true,no

sunny,89,85,false,no

sunny,80,90,true,no

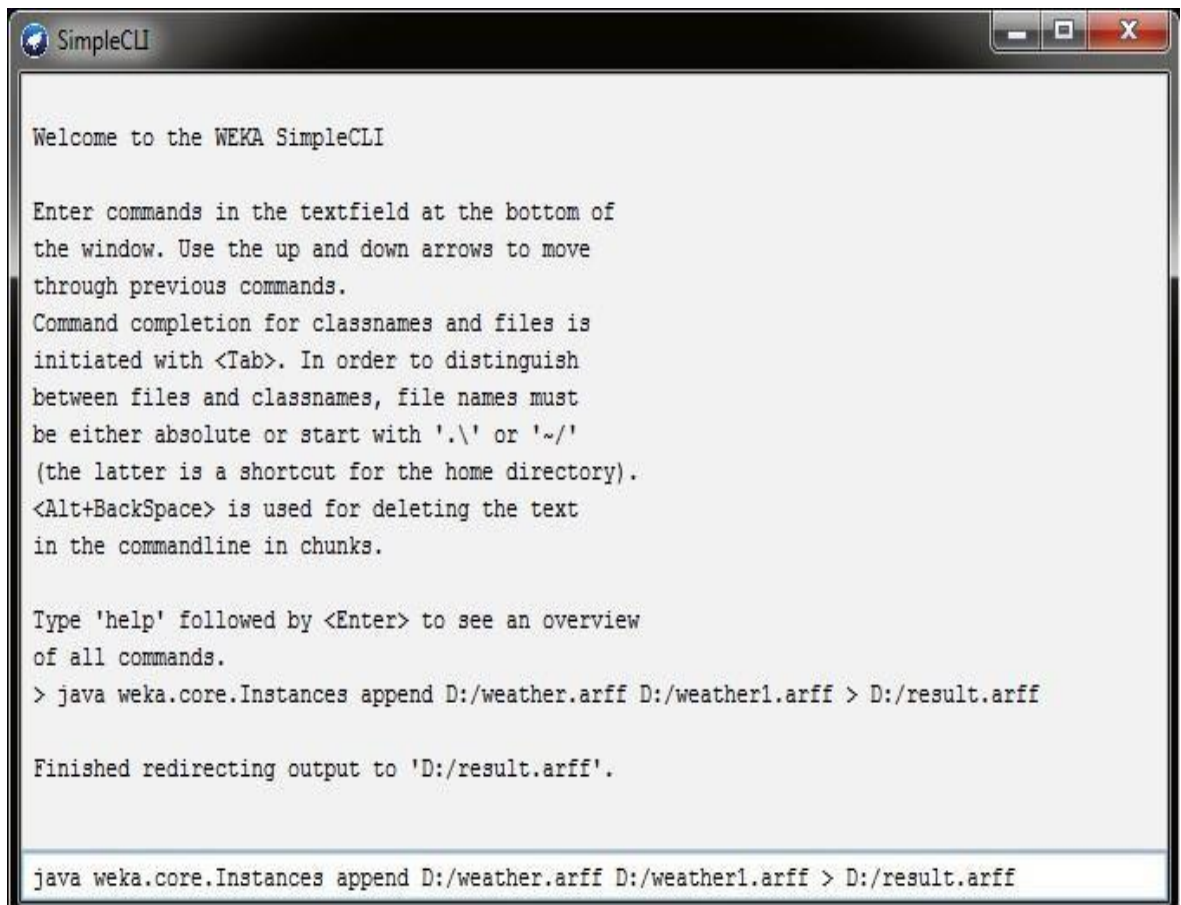
overcast,83,86,false,yes

rainy,67,89,true,yes

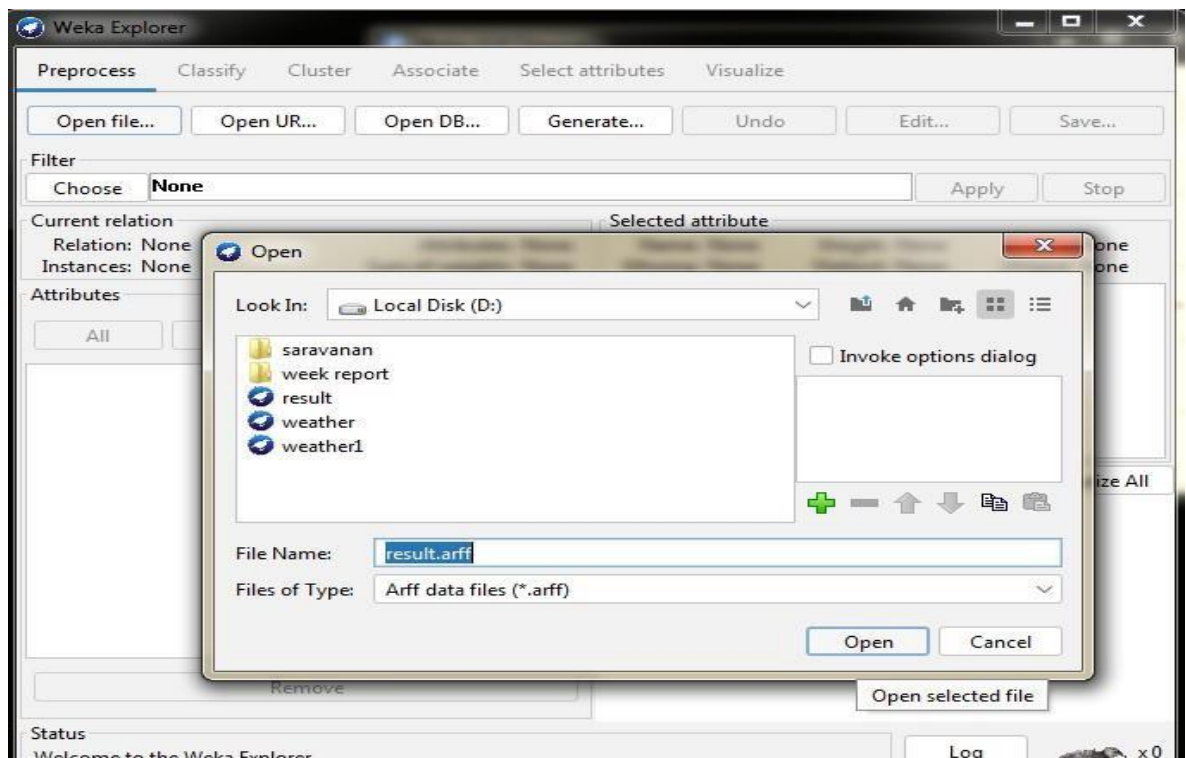
OUTPUT :



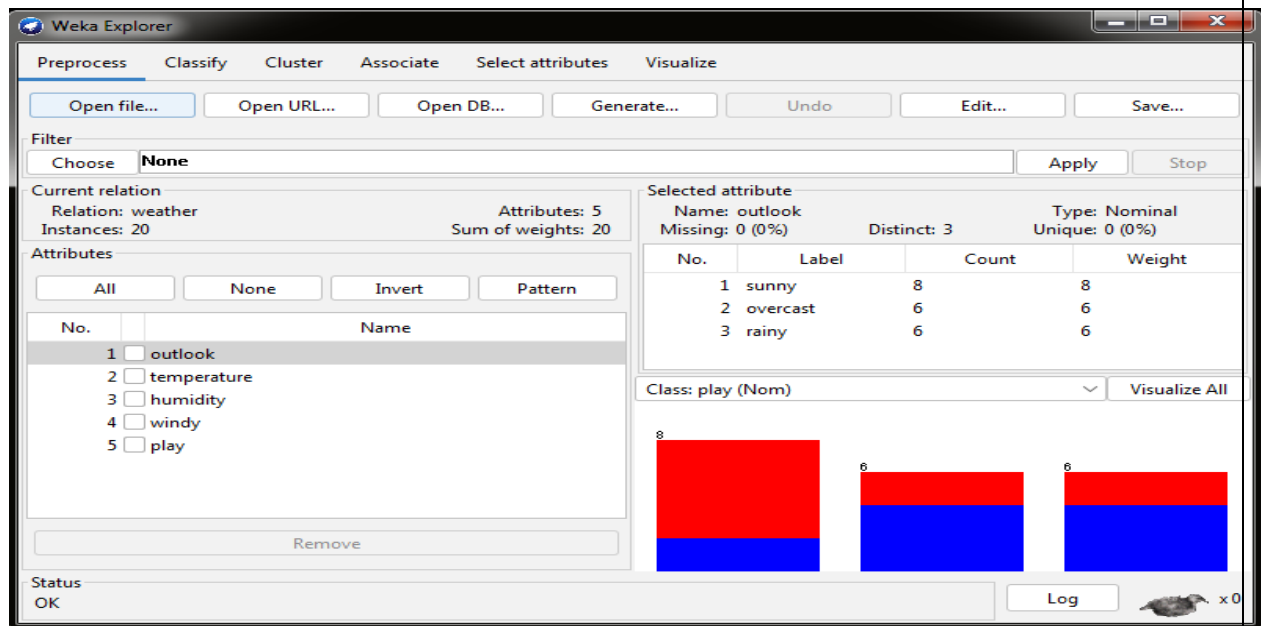
INTEGRATION COMMAND :



DATA EXPLORATION FOR RESULT FILE AFTER INTEGRATION



DATA SET result.arff OUTPUT :



DATASET 1 weather.arff OUTPUT:

Viewer

Relation: weather

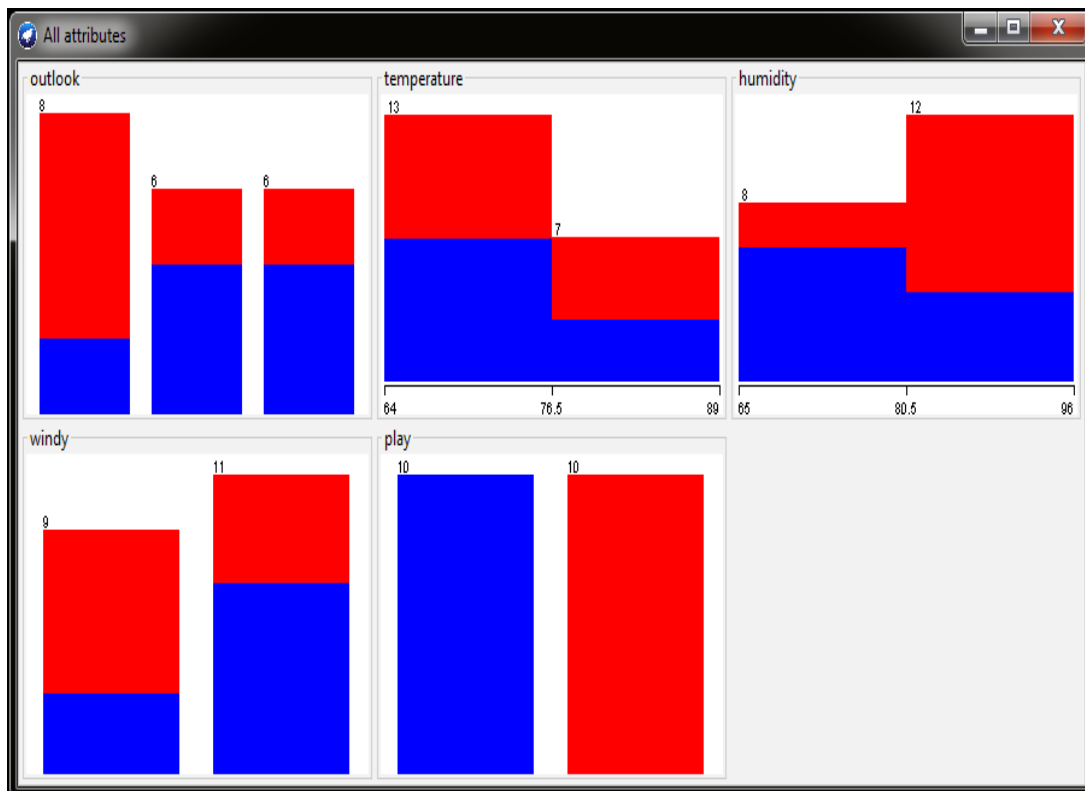
No.	1: outlook Nominal	2: temperature Numeric	3: humidity Numeric	4: windy Nominal	5: play Nominal
1	sunny	85.0	85.0	false	no
2	sunny	80.0	90.0	true	no
3	overcast	83.0	86.0	false	yes
4	rainy	70.0	96.0	false	yes
5	rainy	68.0	80.0	false	yes
6	rainy	65.0	70.0	true	no
7	overcast	64.0	65.0	true	yes
8	sunny	72.0	95.0	false	no
9	sunny	69.0	70.0	false	yes
10	overcast	75.0	90.0	true	no

Buttons: Add instance, Undo, OK, Cancel

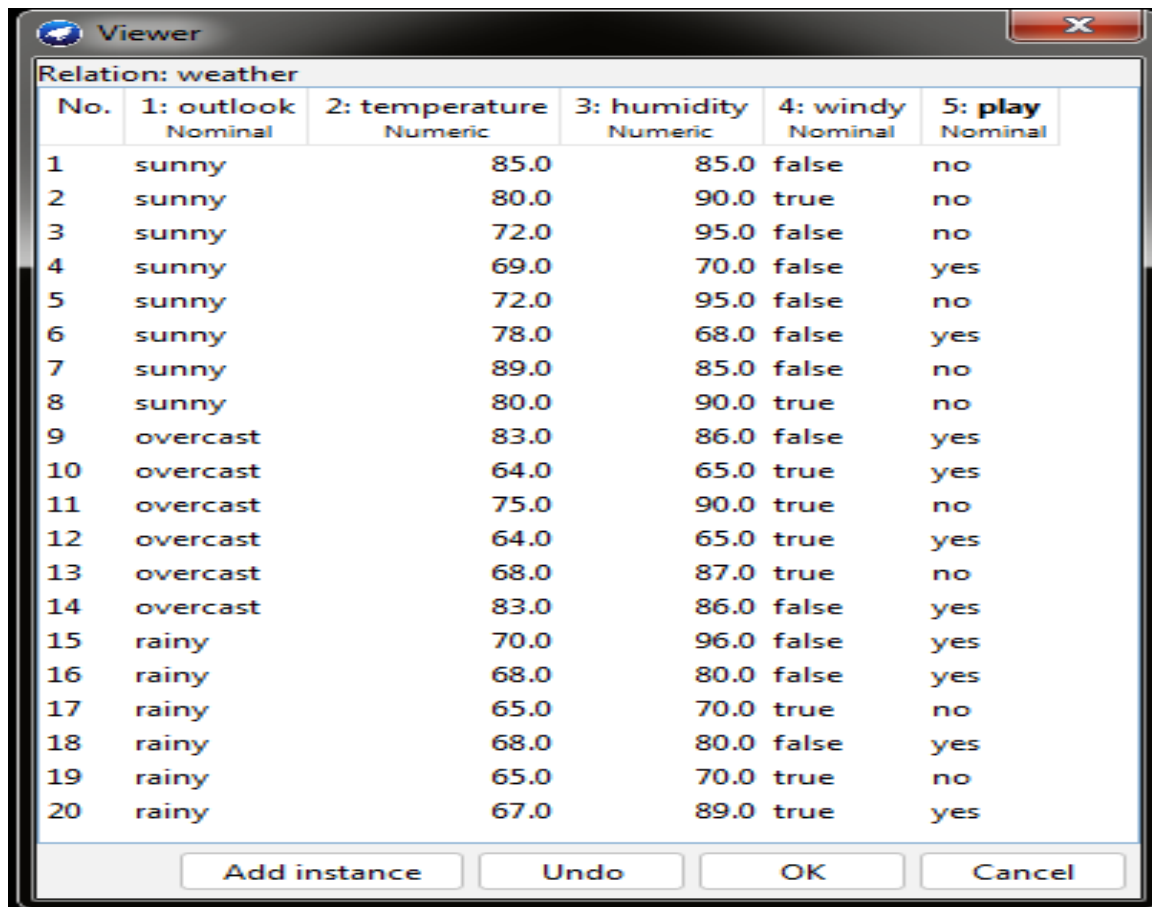
DATASET 2 weather1.arff OUTPUT:

Relation: weather1					
No.	1: outlook Nominal	2: temperature Numeric	3: humidity Numeric	4: windy Nominal	5: play Nominal
1	rainy	68.0	80.0	false	yes
2	rainy	65.0	70.0	true	no
3	overcast	64.0	65.0	true	yes
4	sunny	72.0	95.0	false	no
5	sunny	78.0	68.0	false	yes
6	overcast	68.0	87.0	true	no
7	sunny	89.0	85.0	false	no
8	sunny	80.0	90.0	true	no
9	overcast	83.0	86.0	false	yes
10	rainy	67.0	89.0	true	yes

DATA VISUALIZATION AFTER INTEGRATION :



DATA INTEGRATION :



No.	1: outlook Nominal	2: temperature Numeric	3: humidity Numeric	4: windy Nominal	5: play Nominal
1	sunny	85.0	85.0	false	no
2	sunny	80.0	90.0	true	no
3	sunny	72.0	95.0	false	no
4	sunny	69.0	70.0	false	yes
5	sunny	72.0	95.0	false	no
6	sunny	78.0	68.0	false	yes
7	sunny	89.0	85.0	false	no
8	sunny	80.0	90.0	true	no
9	overcast	83.0	86.0	false	yes
10	overcast	64.0	65.0	true	yes
11	overcast	75.0	90.0	true	no
12	overcast	64.0	65.0	true	yes
13	overcast	68.0	87.0	true	no
14	overcast	83.0	86.0	false	yes
15	rainy	70.0	96.0	false	yes
16	rainy	68.0	80.0	false	yes
17	rainy	65.0	70.0	true	no
18	rainy	68.0	80.0	false	yes
19	rainy	65.0	70.0	true	no
20	rainy	67.0	89.0	true	yes

RESULT :

Thus the program of data exploration and integration with WEKA tool is successfully executed.

Ex.No: 2	APPLY WEKA TOOL FOR DATA VALIDATION
Date:	

AIM:

To perform Data validation of dataset using WEKA tool in Data warehousing .

PROCEDURE :

Data validation using WEKA tool :

Validation -

- Cross-validation, a standard evaluation technique, is a systematic way of running repeated percentage splits.
- Divide a dataset into 10 pieces (“folds”), then hold out each piece in turn for testing and train on the remaining 9 together.
- This gives 10 evaluation results, which are averaged.
- In “stratified” cross-validation, when doing the initial division we ensure that each fold contains approximately the correct proportion of the class values.

To validate the data we are use weather dataset for data validation method .

- For Data validation Open WEKA tool the dialog box displayed on the screen
- Click Explorer → preprocess → open file → weather.arff
- The datas of dataset is explored in the form of current relation , visualization and view table.
- To validate the dataset go to Classify → cross validation → select fold option as 2 to 10 or more → choose any classifiers → click start .
- Result after validation the data shown in classifier output screen.
- Change the classifiers algorithms for multiple methods of validation.

OUTPUT :

DATA VALIDATION :Using ZeroR classifier:

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The 'Classifier' dropdown is set to 'ZeroR'. Under 'Test options', 'Cross-validation' is selected with 'Folds' set to 10. The 'Result list' on the left shows several entries, with '23:29:59 - rules.ZeroR' selected. The 'Classifier output' pane on the right displays the following information:

```
=== Run information ===

Scheme:      weka.classifiers.rules.ZeroR
Relation:    weather
Instances:   10
Attributes:  5
              outlook
              temperature
              humidity
              windy
              play

Test mode:   10-fold cross-validation

=== Classifier model (full training set) ===

ZeroR predicts class value: yes

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      0           0
Incorrectly Classified Instances   10          100
Kappa statistic                    -1
Mean absolute error                 0.5455
Root mean squared error             0.5455
Relative absolute error             100 %
Root relative squared error         100 %
Total Number of Instances          10
```

Using Naïve Bayes classifier:

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The 'Classifier' dropdown is set to 'NaiveBayes'. Under 'Test options', 'Cross-validation' is selected with 'Folds' set to 10. The 'Result list' on the left shows several entries, with '23:39:21 - bayes.NaiveBayes' selected. The 'Classifier output' pane on the right displays the following information:

```
Naive Bayes Classifier

Attribute      Class      yes      no
              (0.5)    (0.5)

outlook
sunny          2.0       4.0
overcast       3.0       2.0
rainy          3.0       2.0
[total]        8.0       8.0

temperature
mean           70.9333   75.1333
std. dev.      7.0155    6.3302
weight sum     5         5
precision      2.3333    2.3333

humidity
mean           79.7143   85.0286
std. dev.      10.8477   7.6192
weight sum     5         5
precision      4.4286    4.4286

windy
true           2.0       4.0
false          5.0       3.0
[total]        7.0       7.0
```

Using BayesNetclassifier :

The screenshot shows the Weka Explorer window with the 'Classify' tab selected. The 'Classifier' dropdown is set to 'BayesNet'. The 'Test options' section shows 'Cross-validation' selected with 'Folds' set to 10. The 'Result list' on the left shows various classifiers, with '23:43:57 - bayes.BayesNet' selected. The 'Classifier output' pane displays the following information:

Bayes Network Classifier
not using ADTree
#attributes=5 #classindex=4
Network structure (nodes followed by parents)
outlook(3): play
temperature(1): play
humidity(1): play
windy(2): play
play(2):
LogScore Bayes: -29.97223259706898
LogScore BDeu: -37.17004963011444
LogScore MDL: -36.675548539429954
LogScore ENTROPY: -28.616500713950785
LogScore AIC: -35.616500713950785

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Metric	Value	Percentage
Correctly Classified Instances	3	30 %
Incorrectly Classified Instances	7	70 %
Kappa statistic	-0.4	
Mean absolute error	0.6378	
Root mean squared error	0.6768	
Relative absolute error	116.9375 %	
Root relative squared error	124.0775 %	
Total Number of Instances	10	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.400	0.800	0.333	0.400	0.164	-0.408	0.240	0.164	yes
	0.200	0.600	0.250	0.200	0.222	-0.408	0.240	0.139	no
Weighted Avg.	0.300	0.700	0.292	0.300	0.293	-0.408	0.240	0.151	

=== Confusion Matrix ===

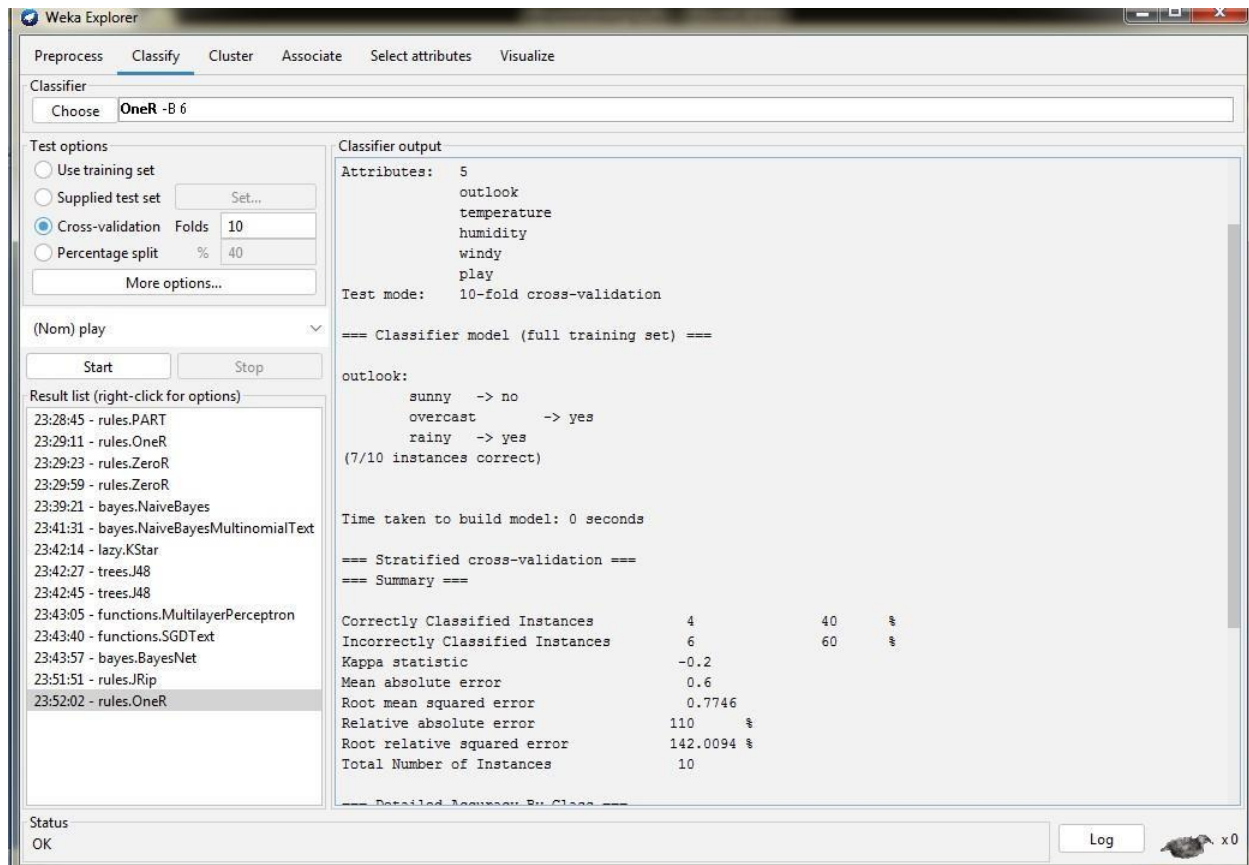
```

a b  <-- classified as
1 3  | a = yes
4 1  | b = no

```

The status bar at the bottom shows 'Status OK' and a 'Log' button.

Using OneRClassifier :



Result :

Thus the dataset data successfully Validated using WEKA tool successfully .

Ex.No: 3	PLAN THE ARCHITECTURE FOR REAL TIME APPLICATION
Date :	

AIM

To plan the **architecture for real time application** in Data warehousing .

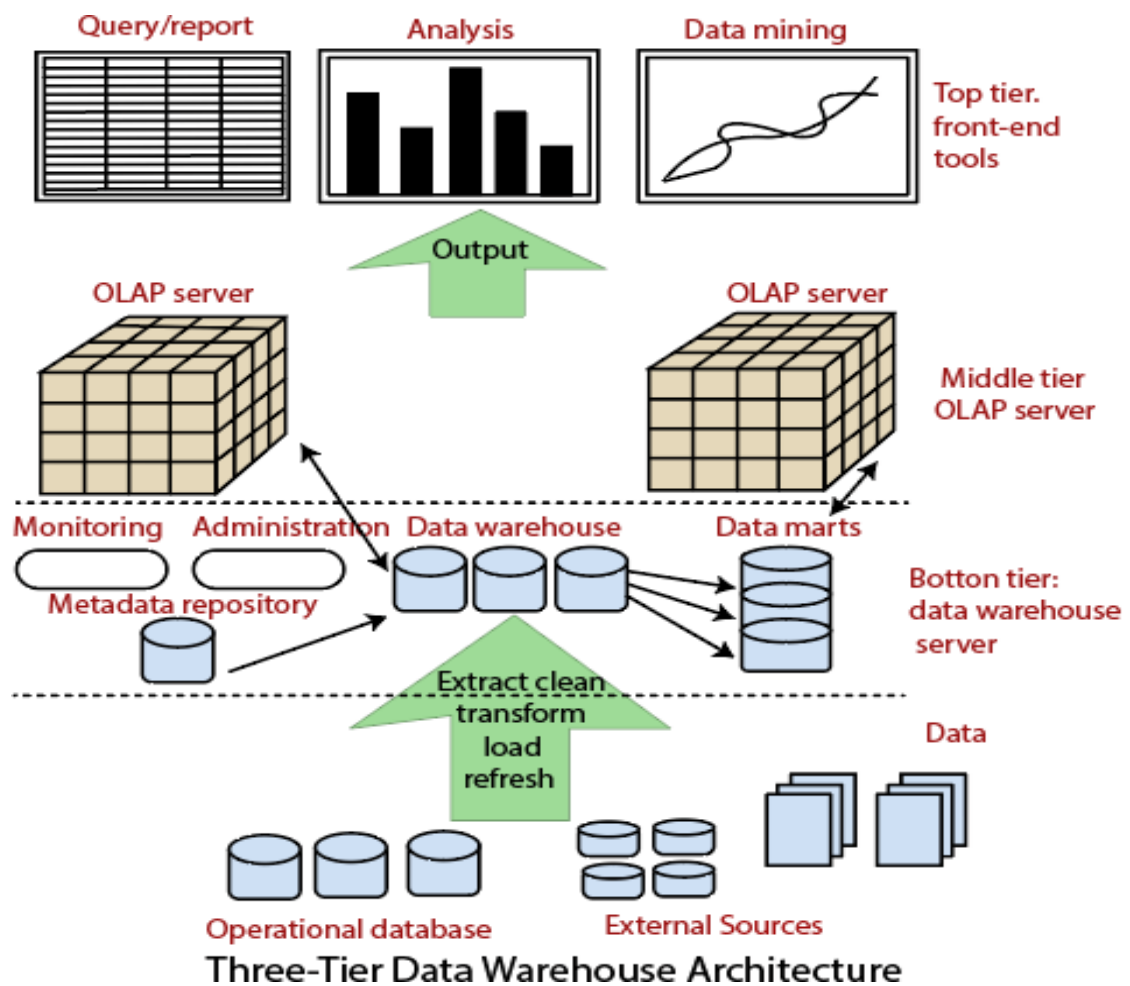
PROCEDURE :

8 steps to data warehouse design:

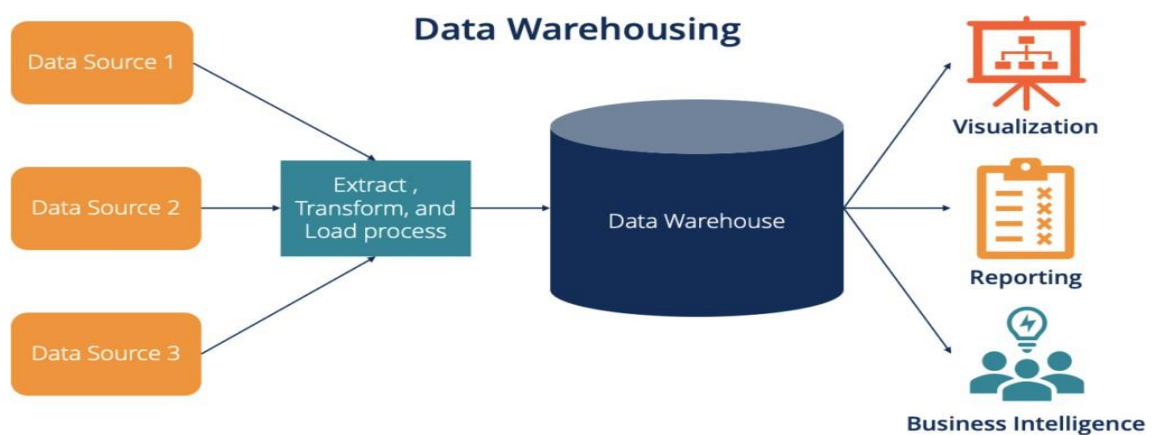
1. **Gather Requirements:** Aligning the business goals and needs of different departments with the overall data warehouse project.
2. **Set Up Environments:** This step is about creating three environments for data warehouse development, testing, and production, each running on separate servers.
3. **Data Modeling:** Design the data warehouse schema, including the fact tables and dimension tables, to support the business requirements.
4. **Develop Your ETL Process:** ETL stands for Extract, Transform, and Load. This process is how data gets moved from its source into your warehouse.
5. **OLAP Cube Design:** Design OLAP cubes to support analysis and reporting requirements.
6. **Reporting & Analysis:** Developing and deploying the reporting and analytics tools that will be used to extract insights and knowledge from the data warehouse.
7. **Optimize Queries:** Optimizing queries ensures that the system can handle large amounts of data and respond quickly to queries.
8. **Establish a Rollout Plan:** Determine how the data warehouse will be introduced to the organization, which groups or individuals will have access to it, and how the data will be presented to these users.

Whether you choose to use a pre-built vendor solution or to start from scratch, you'll need some level of warehouse design to successfully adopt a new data warehouse and get more from your big data.

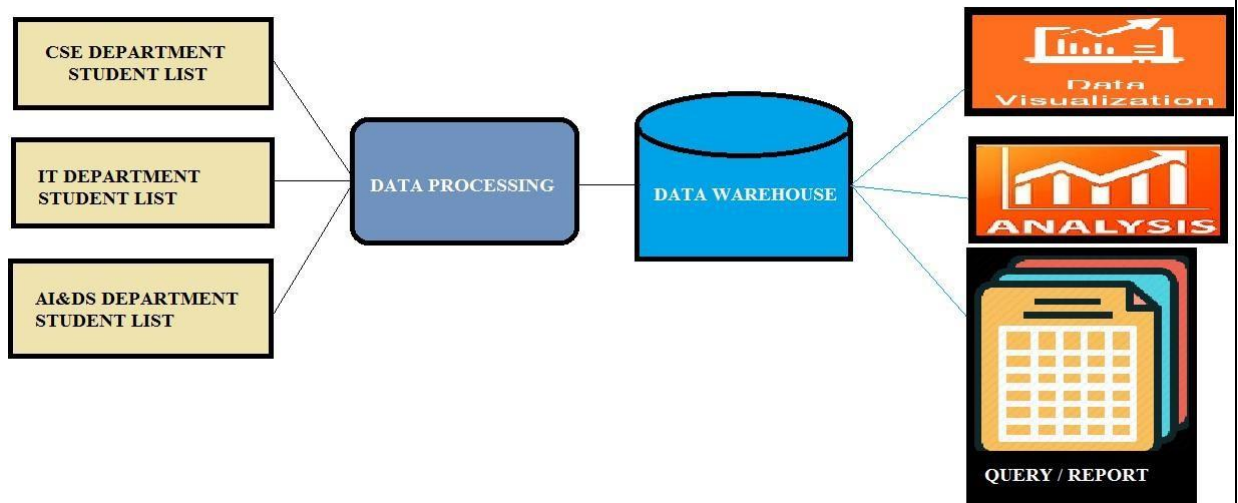
DATA WAREHOUSE THREE TIRE ARCHITECTURE:



ARCHITECTURE FOR REAL TIME APPLICATION :



PLANING ARCHITECTURE FOR STUDENT DATABASE OF IT,CSE,AI&DS DEPARTMENT:



RESULT :

Thus the Architecture for real time application will be planed successfully for Data warehouse real time application.

Ex.No : 4	WRITE A QUERY FOR SCHEMA DEFINITION
Date :	

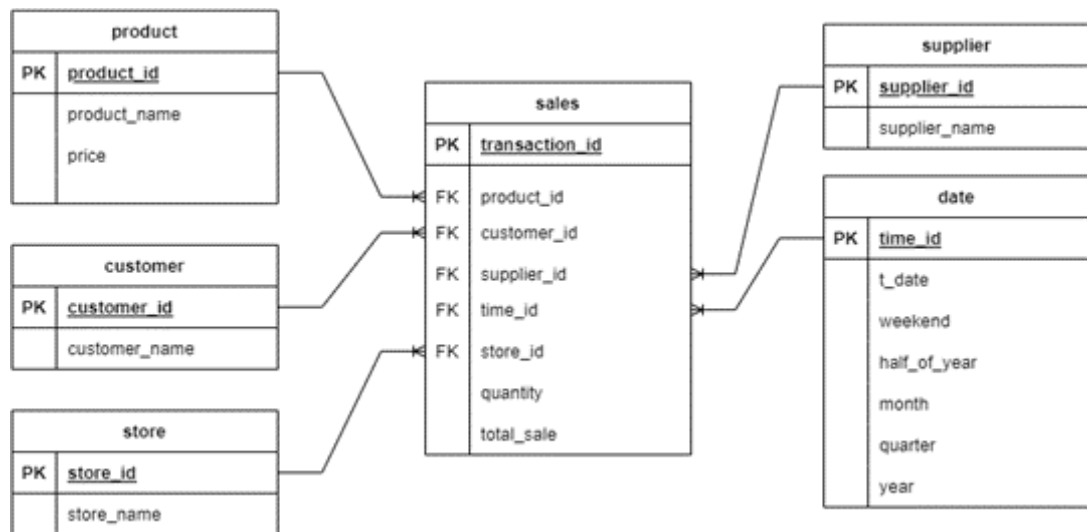
AIM:

To write a query for schema.

PROCEDURE:

Schema Definition: Multidimensional schema is defined using Data Mining Query Language (DMQL). The two primitives, cube definition and dimension definition, can be used for defining the data warehouses and data marts.





Syntax for Cube Definition

```
define cube <cube_name> [ < dimension-list > ]: <measure_list>
```

Syntax for Dimension Definition

```
define dimension <dimension_name> as ( <attribute_or_dimension_list> )
```

Star Schema Definition:

The star schema that we have discussed can be defined using Data Mining Query Language (DMQL) as follows –

```
define cube sales star [time, item, branch, location]:
```

```
dollars sold = sum(sales in dollars), units sold = count(*)
```

```
define dimension time as (time key, day, day of week, month, quarter, year)
```

```
define dimension item as (item key, item name, brand, type, supplier type)
```

```
define dimension branch as (branch key, branch name, branch type)
```

```
define dimension location as (location key, street, city, province or state, country)
```

Snowflake Schema Definition

Snowflake schema can be defined using DMQL as follows –

```
define cube sales snowflake [time, item, branch, location]:
```

```
dollars sold = sum(sales in dollars), units sold = count(*)
```

```
define dimension time as (time key, day, day of week, month, quarter, year)
```

define dimension item as (item key, item name, brand, type, supplier (supplier key, supplier type))

define dimension branch as (branch key, branch name, branch type)

define dimension location as (location key, street, city (city key, city, province or state, country))

Fact Constellation Schema Definition:

Fact constellation schema can be defined using DMQL as follows –

define cube sales [time, item, branch, location]:

dollars sold = sum(sales in dollars), units sold = count(*)

define dimension time as (time key, day, day of week, month, quarter, year)

define dimension item as (item key, item name, brand, type, supplier type)

define dimension branch as (branch key, branch name, branch type)

define dimension location as (location key, street, city, province or state, country)

define cube shipping [time, item, shipper, from location, to location]:

dollars cost = sum(cost in dollars), units shipped = count(*)

define dimension time as time in cube sales

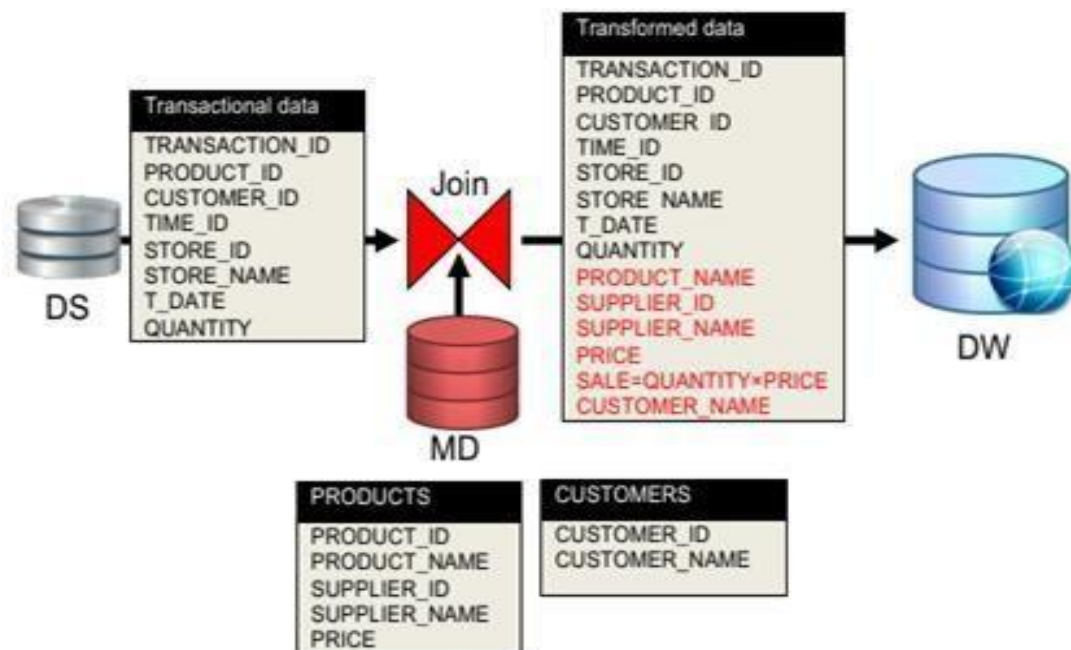
define dimension item as item in cube sales

define dimension shipper as (shipper key, shipper name, location as location in cube sales, shipper type)

define dimension from location as location in cube sales

define dimension to location as location in cube sales

OUTPUT:



Result:

Thus the query using schema definition validate the datawarehouse.

Ex.No : 5	DESIGN THE DATA WAREHOUSE FOR REAL TIME APPLICATION
Date:	

AIM:

To design the real time application on data warehousing using weka.

Procedure:

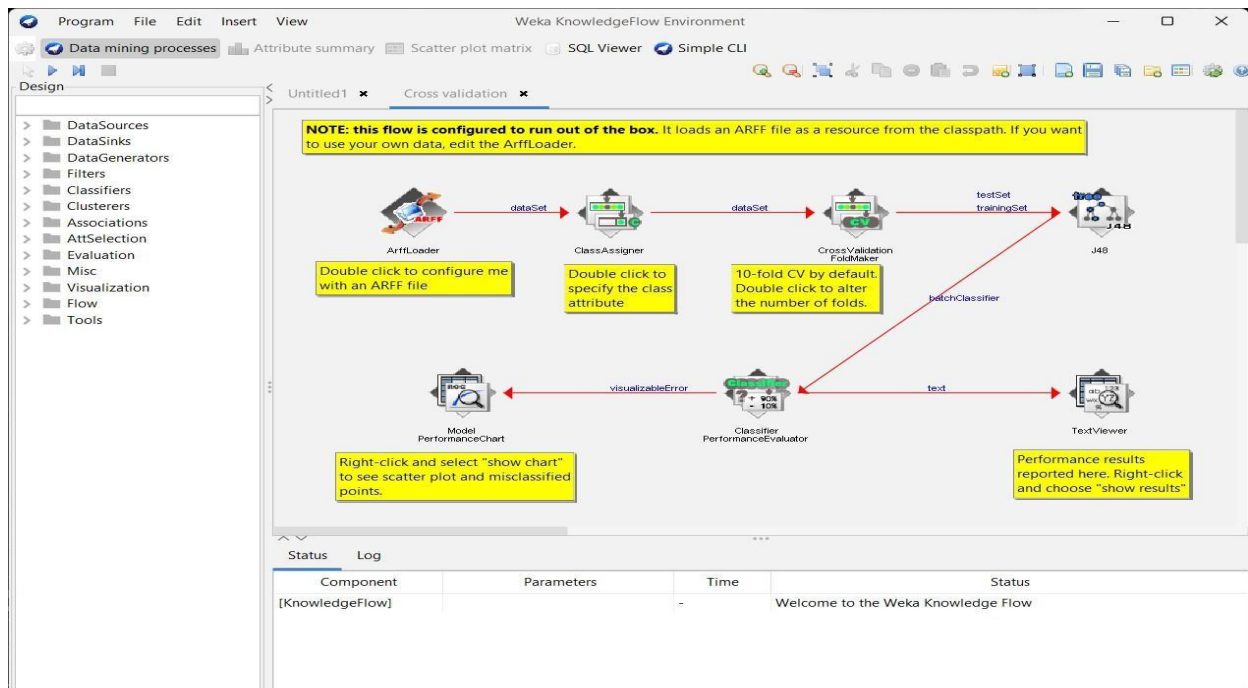
Approach:-

1. Understanding the project requirements
2. Setting up the development environment
3. Implementing the JOIN algorithm
4. Designing the star schema
5. Creating and populating the database
6. Building the near-real-time DW prototype
7. Analyzing the DW prototype
8. Finalizing and presenting

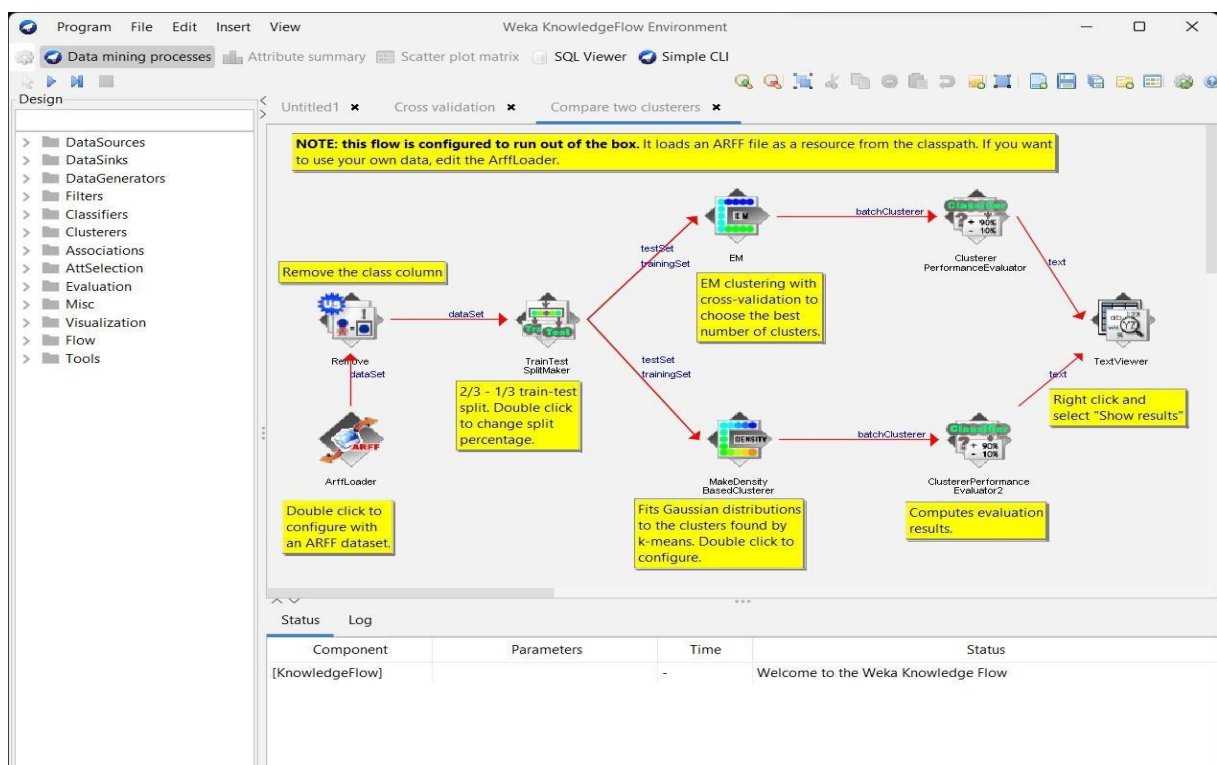
In WEKA TOOL Design data ware house for real time application :

- Open WEKA tool the dialog box displayed on the screen
- Click → Knowledge flow → load a template layout

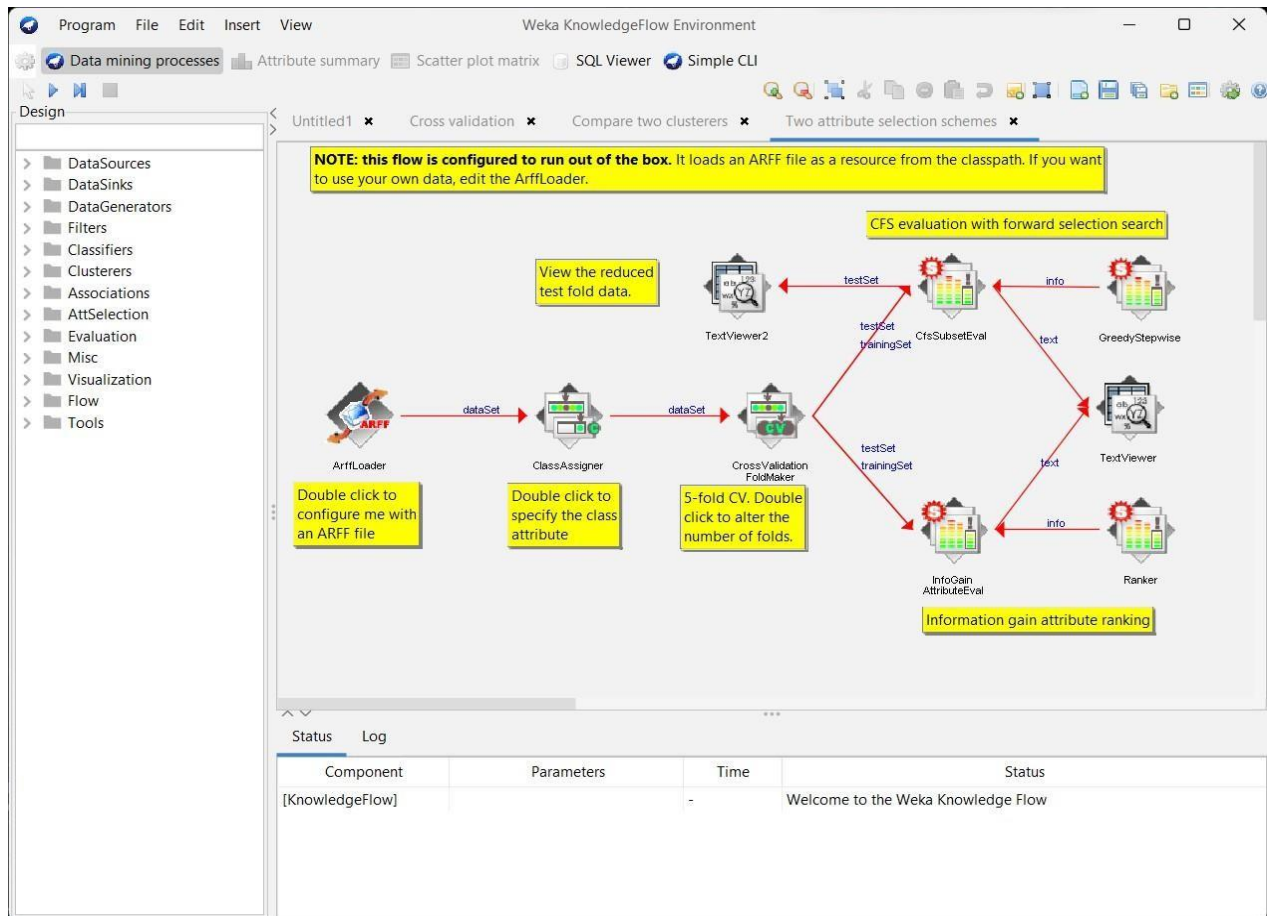
CROSS VALIDATION:



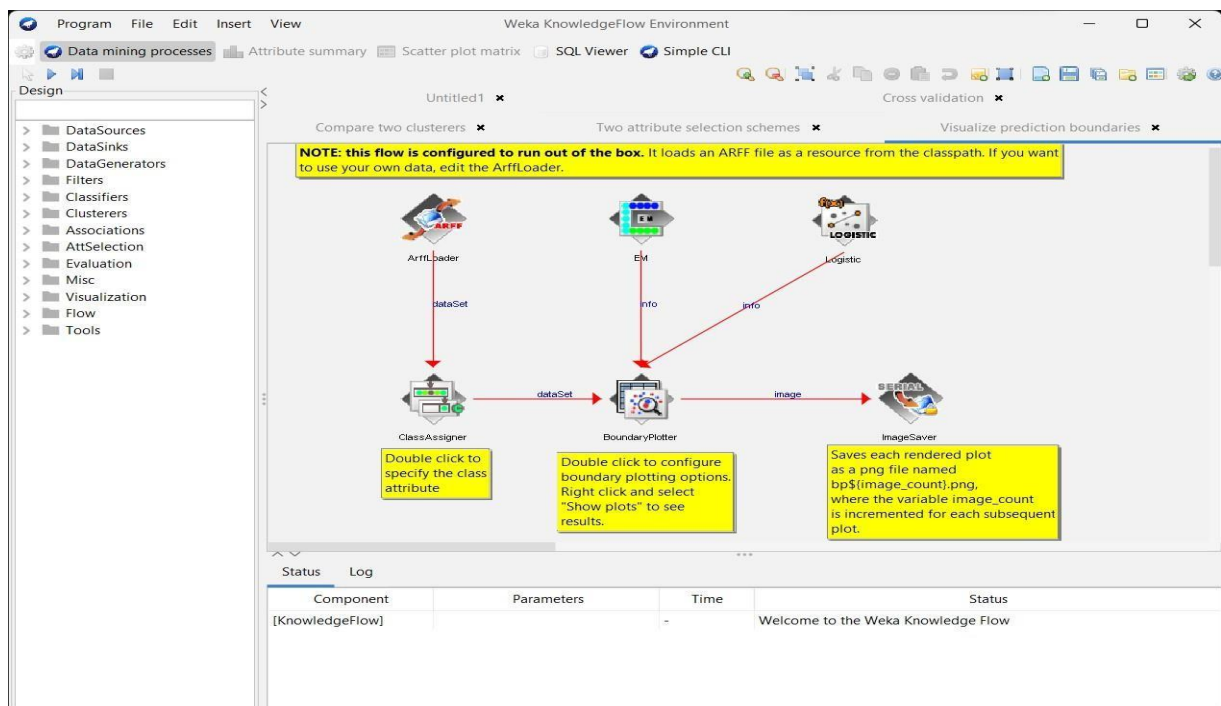
COMPARE TWO CLUSTERS:



TWO ATTRIBUTE SELECTION SCHEMES:



VISUALIZE PREDICTION BOUNDARIES:



Result:

The successfully build and analysed a real-time Data Warehouse prototype.

Ex.No : 6	ANALYSE THE DIMENSIONAL MODELING
Date :	

AIM:

To Analyse the dimensional modeling using WEKA tool.

Procedure:

- In short the goal of dimensional modelling can be summarized as,
- In the following examples we will choose a practical business scenario and see how to identify dimensions and facts to model the scenario

Step by Step Approach to Dimensional Modeling

Consider the business scenario for a fastfood chain below

- The business objective is to create a data model that can store and report number of burgers and fries sold from a specific McDonalds outlet per day.

Step 1: Identify the dimensions

Step 2: Identify the measures

Step 3: Identify the attributes or properties of dimensions

Step 4: Identify the granularity of the measures

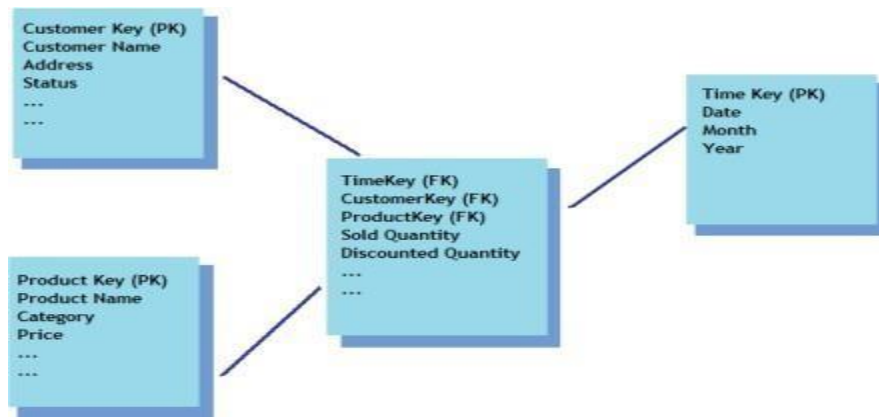
Step 5: History Preservation (Optional)

Food

KEY	NAME
1	Burger
2	Fries

Store

KEY	NAME
1	Store 1
2	Store 2
3	Store ...



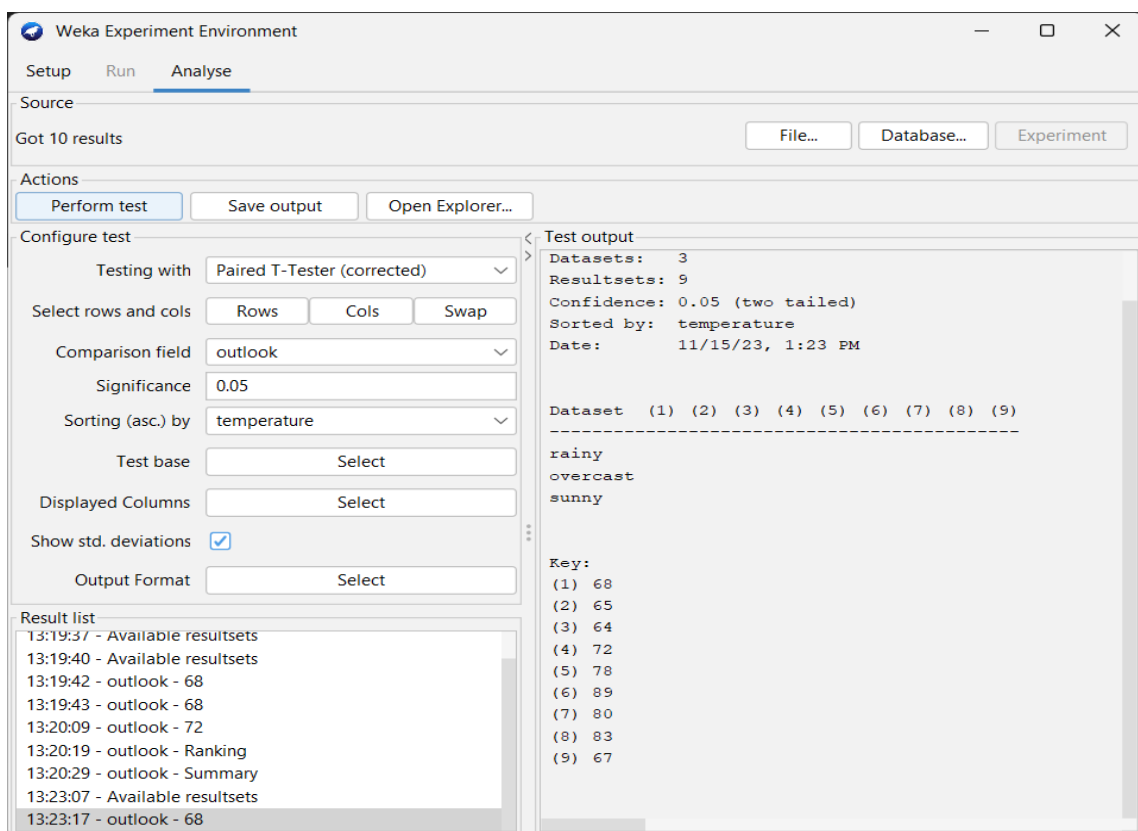
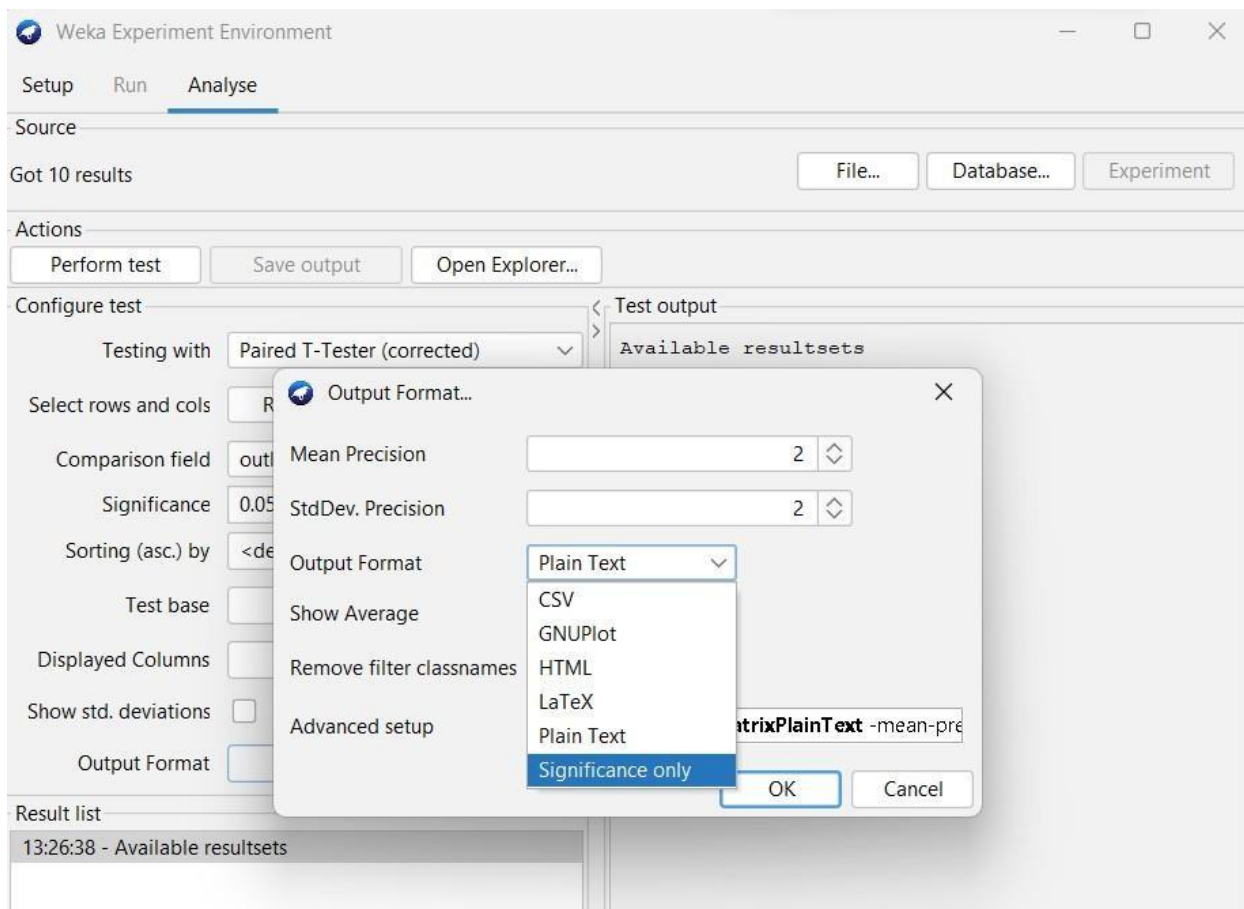
Analyse the Dimensional model using WEKA tool

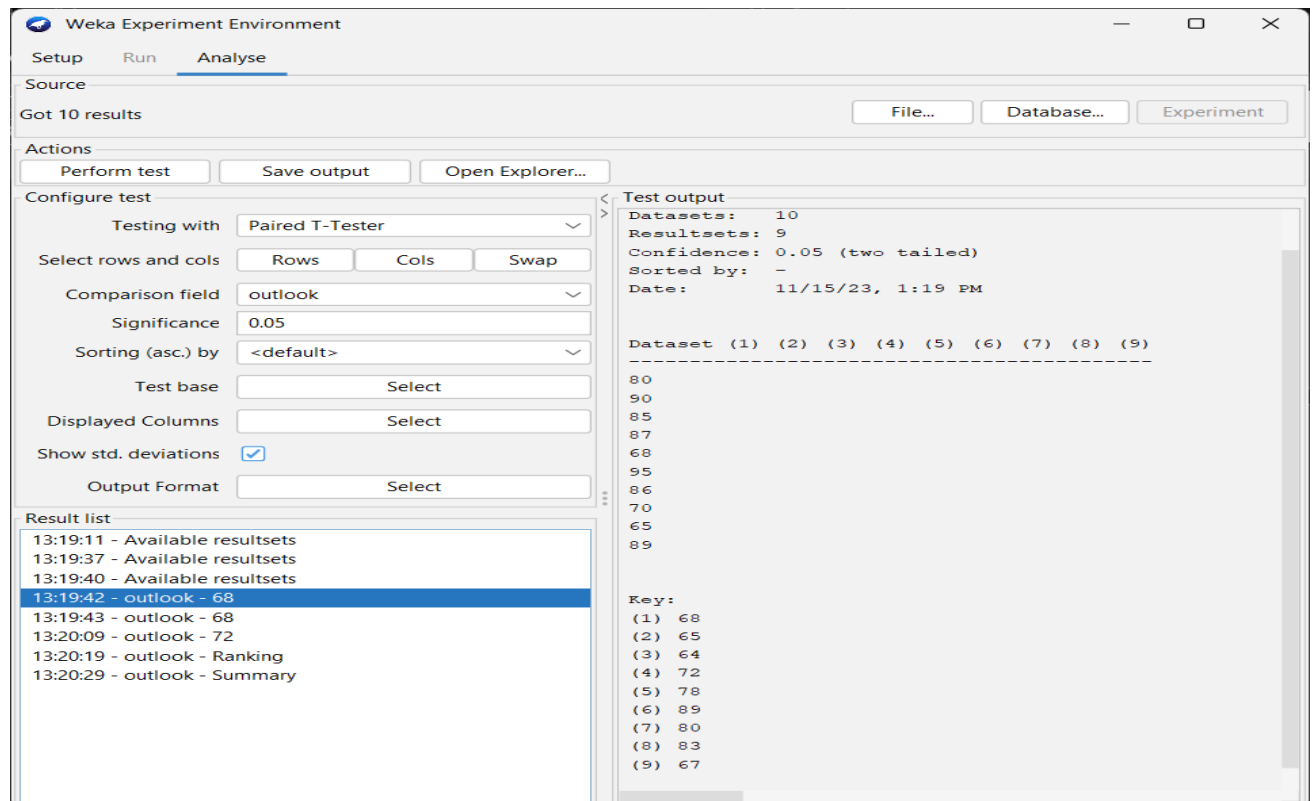
- Open WEKA tool the dialog box displayed on the screen
- Click Experimenter → Analyse
- In Analyse it shows many options. select " file " option and open the file with .arff file formate.
- To analyse the datas go to select rows and cols for selecting rows and coloums option select the datas we need to analyse.
- Choose the comparison field attribute that need to analyse
- Select the sorting option, configuration testing, test base, displayed columns, standard deviations, output format for analyse the data set.
- After choose the needed options to perform the analyse data click Perform test.

OUTPUT:



Analyse the dataset:





Result:

The program successfully build and analyze the dimensional modeling.

Ex.No: 7	CASE STUDY USING OLAP
Date:	

AIM:

To Study about Online Analytical Processing.

CASE STUDY:

OLAP (Online Analytical Processing)

OLAP stands for **On-Line Analytical Processing**. OLAP is a classification of software technology which authorizes analysts, managers, and executives to gain insight into information through fast, consistent, interactive access in a wide variety of possible views of data that has been transformed from raw information to reflect the real dimensionality of the enterprise as understood by the clients.

Uses of OLAP

Finance and accounting:

- Budgeting
- Activity-based costing
- Financial performance analysis
- And financial modeling

Sales and Marketing

- Sales analysis and forecasting
- Market research analysis
- Promotion analysis
- Customer analysis
- Market and customer segmentation

Production

- Production planning
- Defect analysis

OLAP cubes have two main purposes. The first is to provide business users with a data model more intuitive to them than a tabular model. This model is called a Dimensional Model.

The second purpose is to enable fast query response that is usually difficult to achieve using tabular models.



1) Multidimensional Conceptual View: This is the central features of an OLAP system. By needing a multidimensional view, it is possible to carry out methods like slice and dice.

2) Transparency: Make the technology, underlying information repository, computing operations, and the dissimilar nature of source data totally transparent to users. Such transparency helps to improve the efficiency and productivity of the users.

3) Accessibility: It provides access only to the data that is actually required to perform the particular analysis, present a single, coherent, and consistent view to the clients. The OLAP system must map its own logical schema to the heterogeneous physical data stores and perform any necessary transformations. The OLAP operations should be sitting between data sources (e.g., data warehouses) and an OLAP front-end.

4) Consistent Reporting Performance: To make sure that the users do not feel any significant degradation in documenting performance as the number of dimensions or the size of the database increases. That is, the performance of OLAP should not suffer as the number of dimensions is increased. Users must observe consistent run time, response time, or machine utilization every time a given query is run.

5) Client/Server Architecture: Make the server component of OLAP tools sufficiently intelligent that the various clients to be attached with a minimum of effort and integration programming. The server should be capable of mapping and consolidating data between dissimilar databases.

6) Generic Dimensionality: An OLAP method should treat each dimension as equivalent in both its structure and operational capabilities. Additional operational capabilities may be allowed to selected dimensions, but such additional tasks should be grantable to any dimension.

7) Dynamic Sparse Matrix Handling: To adapt the physical schema to the specific analytical model being created and loaded that optimizes sparse matrix handling. When encountering the sparse matrix, the system must be easy to dynamically assume the distribution of the information and adjust the storage and access to obtain and maintain a consistent level of performance.

8) Multiuser Support: OLAP tools must provide concurrent data access, data integrity, and access security.

9) Unrestricted cross-dimensional Operations: It provides the ability for the methods to identify dimensional order and necessarily functions roll-up and drill-down methods within a dimension or across the dimension.

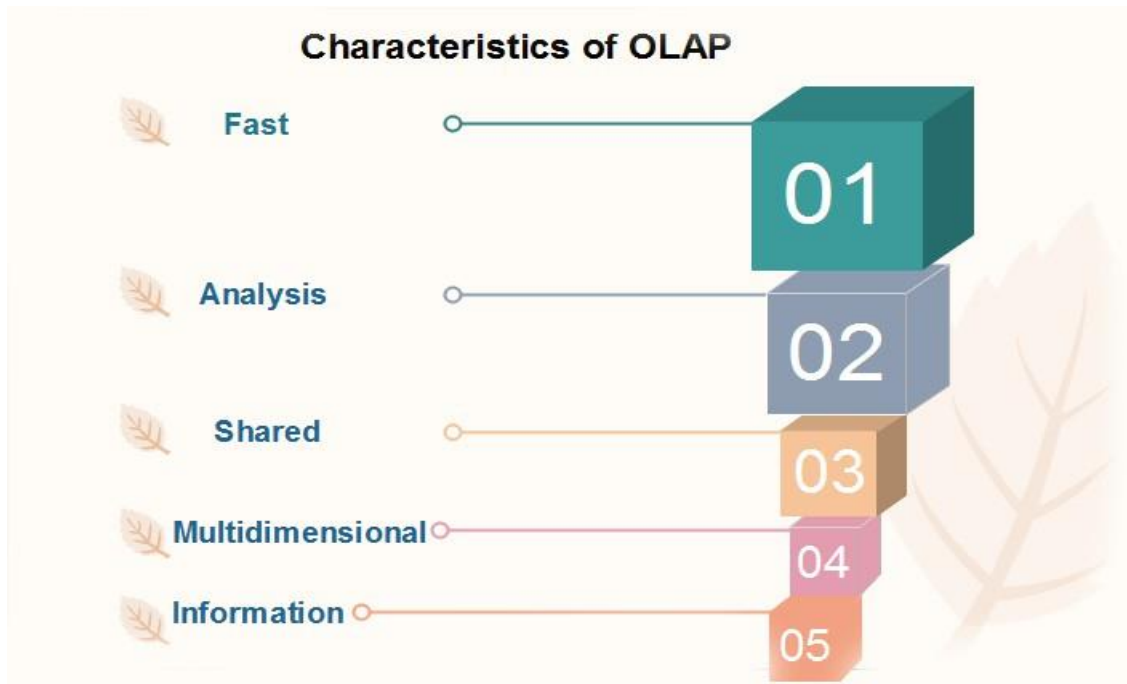
10) Intuitive Data Manipulation: Data Manipulation fundamental the consolidation direction like as reorientation (pivoting), drill-down and roll-up, and another manipulation to be accomplished naturally and precisely via point-and-click and drag and drop methods on the cells of the scientific model. It avoids the use of a menu or multiple trips to a user interface.

11) Flexible Reporting: It implements efficiency to the business clients to organize columns, rows, and cells in a manner that facilitates simple manipulation, analysis, and synthesis of data.

12) Unlimited Dimensions and Aggregation Levels: The number of data dimensions should be unlimited. Each of these common dimensions must allow a practically unlimited number of customer-defined aggregation levels within any given consolidation path.

Characteristics of OLAP

In the **FASMI characteristics of OLAP methods**, the term derived from the first letters of the characteristics are:



Fast

It defines which the system targeted to deliver the most feedback to the client within about five seconds, with the elementary analysis taking no more than one second and very few taking more than 20 seconds.

Analysis

It defines which the method can cope with any business logic and statistical analysis that is relevant for the function and the user, keep it easy enough for the target client. Although some preprogramming may be needed we do not think it acceptable if all application definitions have to be allow the user to define new Adhoc calculations as part of the analysis and to document on the data in any desired method, without having to program so we excludes products (like Oracle Discoverer) that do not allow the user to define new Adhoc calculation as part of the analysis and to document on the data in any desired product that do not allow adequate end user-oriented calculation flexibility.

Share

It defines which the system tools all the security requirements for understanding and, if multiple write connection is needed, concurrent update location at an appropriated level, not all functions need customer to write data back, but for the increasing number which does, the system should be able to manage multiple updates in a timely, secure manner.

Multidimensional

This is the basic requirement. OLAP system must provide a multidimensional conceptual view of the data, including full support for hierarchies, as this is certainly the most logical method to analyze business and organizations.

Information

The system should be able to hold all the data needed by the applications. Data sparsity should be handled in an efficient manner.

The main characteristics of OLAP are as follows:

1. **Multidimensional conceptual view:** OLAP systems let business users have a dimensional and logical view of the data in the data warehouse. It helps in carrying slice and dice operations.
2. **Multi-User Support:** Since the OLAP techniques are shared, the OLAP operation should provide normal database operations, containing retrieval, update, adequacy control, integrity, and security.
3. **Accessibility:** OLAP acts as a mediator between data warehouses and front-end. The OLAP operations should be sitting between data sources (e.g., data warehouses) and an OLAP front-end.
4. **Storing OLAP results:** OLAP results are kept separate from data sources.
5. **Uniform documenting performance:** Increasing the number of dimensions or database size should not significantly degrade the reporting performance of the OLAP system.
6. OLAP provides for distinguishing between zero values and missing values so that aggregates are computed correctly.
7. OLAP system should ignore all missing values and compute correct aggregate values.
8. OLAP facilitate interactive query and complex analysis for the users.
9. OLAP allows users to drill down for greater details or roll up for aggregations of metrics along a single business dimension or across multiple dimension.
10. OLAP provides the ability to perform intricate calculations and comparisons.
11. OLAP presents results in a number of meaningful ways, including charts and graphs.

Result:

Thus the case study about OLAP was successfully implemented.

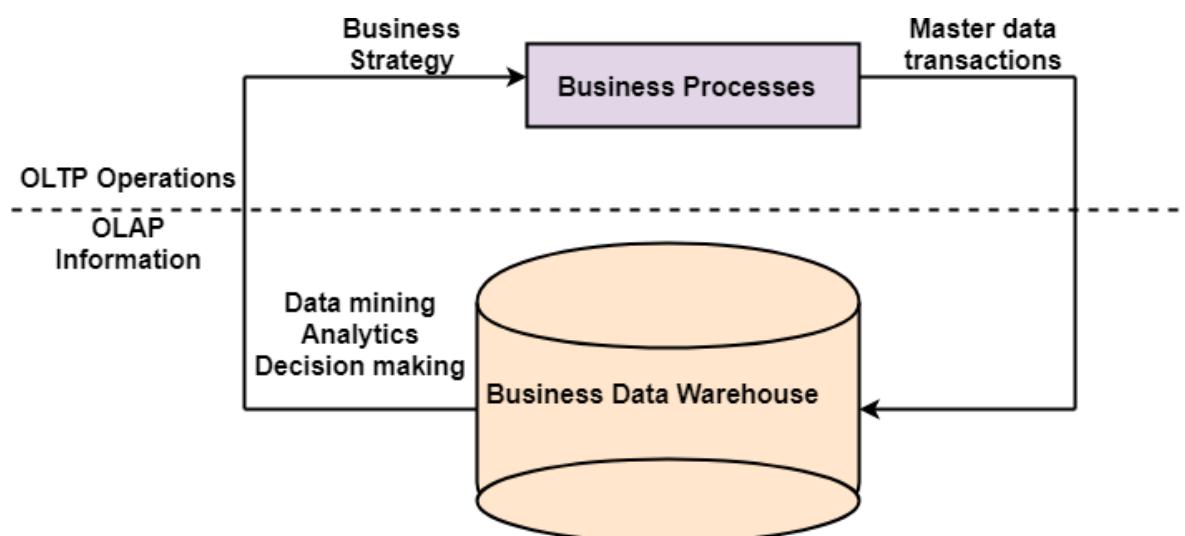
Ex.No : 8	CASE STUDY USING OTLP
Date:	

AIM:

To Study about Online Transaction Processing.

CASE STUDY:

OLTP (On-Line Transaction Processing) is featured by a large number of short on-line transactions (INSERT, UPDATE, and DELETE). The primary significance of OLTP operations is put on very rapid query processing, maintaining record integrity in multi-access environments, and effectiveness consistent by the number of transactions per second. In the OLTP database, there is an accurate and current record, and schema used to save transactional database is the entity model (usually 3NF).



1) Users: **OLTP** systems are designed for office worker while the **OLAP** systems are designed for decision-makers. Therefore while an **OLTP** method may be accessed by hundreds or even thousands of clients in a huge enterprise, an **OLAP** system is suitable to be accessed only by a select class of manager and may be used only by dozens of users.

2) Functions: **OLTP** systems are mission-critical. They provide day-to-day operations of an enterprise and are largely performance and availability driven. These operations carry out simple repetitive operations. **OLAP** systems are management-critical to support the decision of enterprise support tasks using detailed investigation.

3) Nature: Although **SQL** queries return a set of data, **OLTP** methods are designed to step one record at the time, for example, a data related to the user who may be on the phone or in the store. **OLAP** system is not designed to deal with individual customer records. Instead, they include queries that deal with many data at a time and provide summary or aggregate information to a manager. **OLAP** applications include data stored in a data warehouses that have been extracted from many tables and possibly from more than one enterprise database.

4) Design: **OLTP** database operations are designed to be application-oriented while **OLAP** operations are designed to be subject-oriented. OLTP systems view the enterprise record as a collection of tables (possibly based on an entity-relationship model). **OLAP** operations view enterprise information as multidimensional).

5) Data: OLTP systems usually deal only with the current status of data. For example, a record about an employee who left three years ago may not be feasible on the Human Resources System. The old data may have been achieved on some type of stable storage media and may not be accessible online. On the other hand, OLAP systems needed historical data over several years since trends are often essential in decision making.

6) Kind of use: **OLTP** methods are used for reading and writing operations while OLAP methods usually do not update the data.

7) View: An **OLTP** system focuses primarily on the current data within an enterprise or department, which does not refer to historical data or data in various organizations. In contrast, an **OLAP** system spans multiple version of a database schema, due to the evolutionary process of an organization. OLAP system also deals with information that originates from different organizations, integrating information from many data stores. Because of their huge volume, these are stored on multiple storage media.

8) Access Patterns: The access pattern of an OLTP system consist primarily of short, atomic transactions. Such a system needed concurrency control and recovery techniques. However, access to OLAP systems is mostly read-only operations because these data warehouses store historical information.

Result:

Thus the case study about OLTP was successfully implemented.

Ex.No : 9	IMPLEMENTATION OF WAREHOUSE TESTING
Date :	

AIM :

To perform Data Warehouse testing perform the Data exploration ,integration, Data validation ,Data analyse , Visualizing dataset using WEKA tool in Data warehousing .

PROCEDURE:

- Open start → programs → Accessories → Notepad++
- Type the following sample dataset program on Notepad++ for creating Weather table.
- After the weather table program created save the file name in .arff(attribute-relation file format) file formate .
- For Data exploration Open WEKA tool the dialog box displayed on the screen
- Click Explorer → preprocess
- In Preprocess it shows many options. select " open file " option and open the file with .arff file formate.
- The attributes of our program displayed on the screen with current relation , visualize all data's.
- To view the table go to Edit option the viewer shows the table with attributes and datas.

PROGRAM :

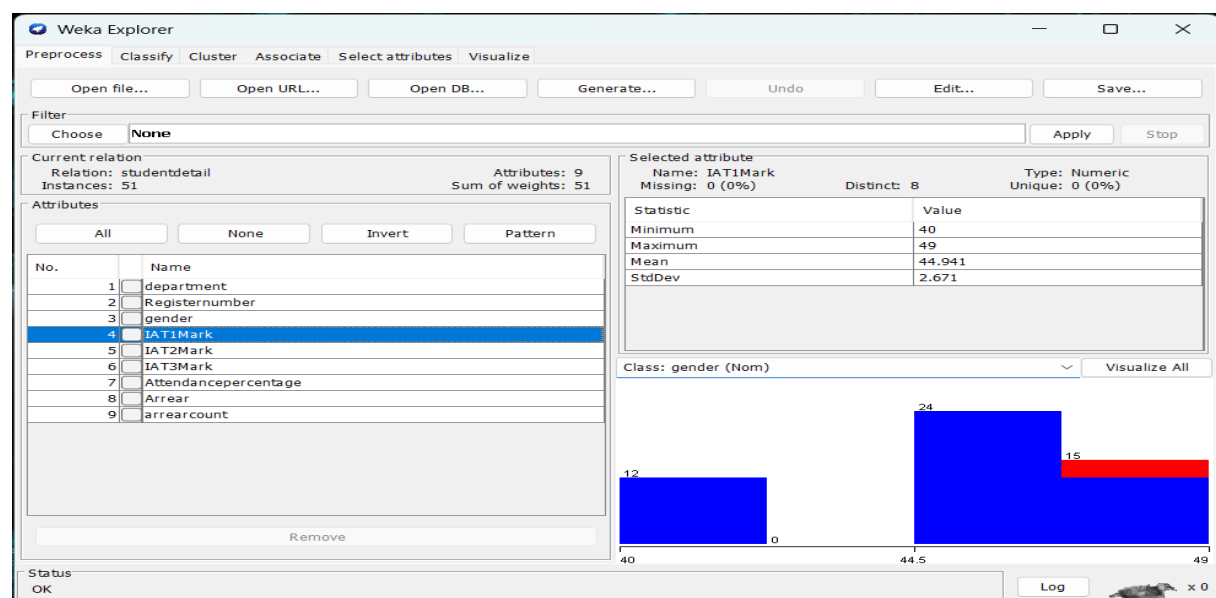
```
@relation studentdetail
@attribute department {CSE,IT}
@attribute Registernumber numeric
@attribute gender {M,F,O}
@attribute IAT1Mark numeric
@attribute IAT2Mark numeric
@attribute IAT3Mark numeric
@attribute Attendancepercentage numeric
@attribute Arrear {yes,no}
@attribute arrearcount numeric
```

@data

CSE,620821104002,M,45,46,45,98,no,0
CSE,620821104003,M,46,47,49,95,no,0
CSE,620821104004,M,47,42,45,90,yes,2
CSE,620821104005,M,40,47,48,93,yes,1
CSE,620821104006,M,42,41,47,98,yes,1
CSE,620821104007,M,45,46,49,100,yes,2
CSE,620821104008,M,48,46,48,90,no,0
CSE,620821104011,M,46,41,43,95,no,0
CSE,620821104012,M,41,43,45,98,no,0
IT,620821104071,M,47,46,48,99,no,0
IT,620821104072,M,45,43,43,90,no,0
IT,620821104073,M,45,46,48,98,yes,0
IT,620821104074,M,45,46,47,90,no,0
IT,620821104075,M,40,,43,45,89,yes,0
IT,620821104076,F,49,44,44,98,no,0
IT,620821104077,M,45,49,44,98,no,0
IT,620821104078,M,48,45,47,89,no,0

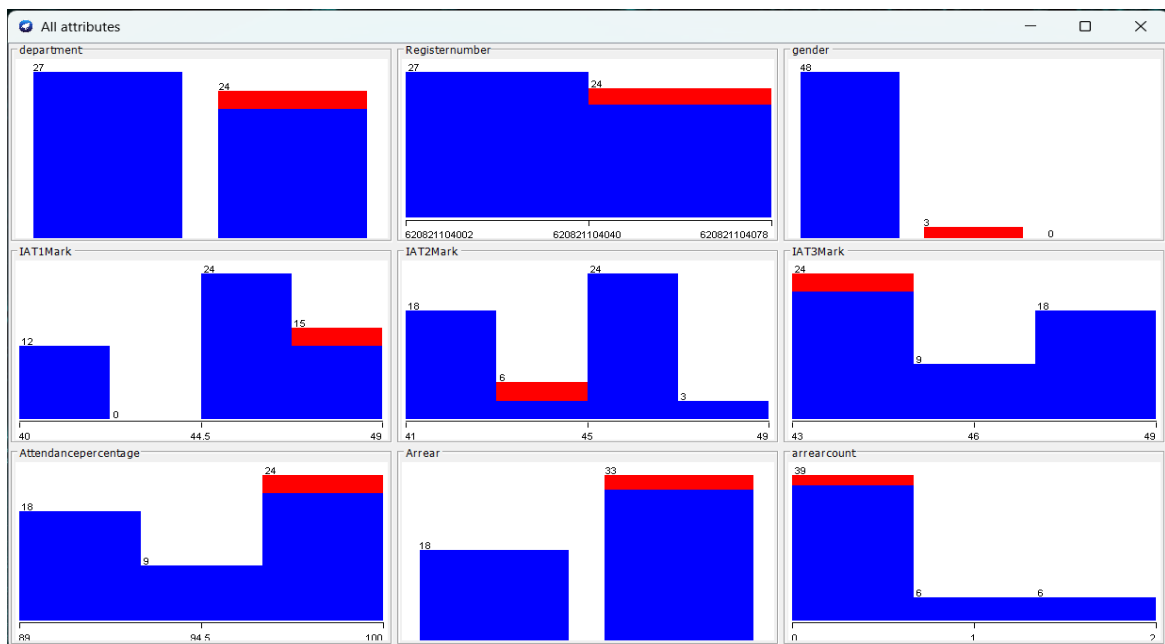
OUTPUT:

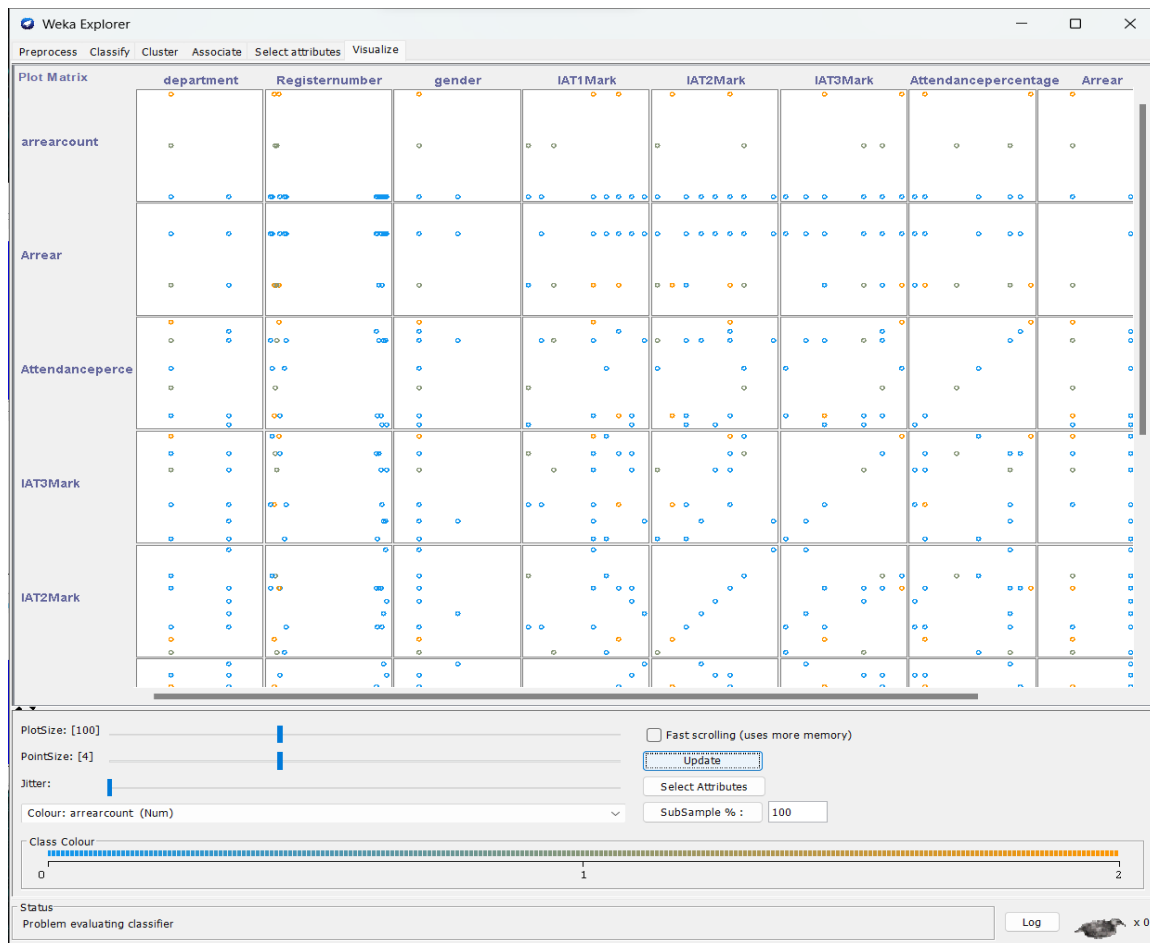
DATA EXPLORATION OF DATA WAREHOUSE:



Viewer									
Relation: studentdetail									
No.	1: department Nominal	2: Registernumber Numeric	3: gender Nominal	4: IAT1Mark Numeric	5: IAT2Mark Numeric	6: IAT3Mark Numeric	7: Attendancepercentage Numeric	8: Arrear Nominal	9: arrearcount Numeric
1	CSE	6.20821104002E11	M	45.0	46.0	45.0	98.0	no	0.0
2	CSE	6.20821104003E11	M	46.0	47.0	49.0	95.0	no	0.0
3	CSE	6.20821104004E11	M	47.0	42.0	45.0	90.0	yes	2.0
4	CSE	6.20821104005E11	M	40.0	47.0	48.0	93.0	yes	1.0
5	CSE	6.20821104006E11	M	42.0	41.0	47.0	98.0	yes	1.0
6	CSE	6.20821104007E11	M	45.0	46.0	49.0	100.0	yes	2.0
7	CSE	6.20821104008E11	M	48.0	46.0	48.0	90.0	no	0.0
8	CSE	6.20821104011E11	M	46.0	41.0	43.0	95.0	no	0.0
9	CSE	6.20821104012E11	M	41.0	43.0	45.0	98.0	no	0.0
10	IT	6.20821104071E11	M	47.0	46.0	48.0	99.0	no	0.0
11	IT	6.20821104072E11	M	45.0	43.0	43.0	90.0	no	0.0
12	IT	6.20821104073E11	M	45.0	46.0	48.0	98.0	yes	0.0
13	IT	6.20821104074E11	M	45.0	46.0	47.0	90.0	no	0.0
14	IT	6.20821104075E11	M	40.0	43.0	45.0	89.0	yes	0.0
15	IT	6.20821104076E11	F	49.0	44.0	44.0	98.0	no	0.0
16	IT	6.20821104077E11	M	45.0	49.0	44.0	98.0	no	0.0
17	IT	6.20821104078E11	M	48.0	45.0	47.0	89.0	no	0.0
18	CSE	6.20821104002E11	M	45.0	46.0	45.0	98.0	no	0.0
19	CSE	6.20821104003E11	M	46.0	47.0	49.0	95.0	no	0.0
20	CSE	6.20821104004E11	M	47.0	42.0	45.0	90.0	yes	2.0
21	CSE	6.20821104005E11	M	40.0	47.0	48.0	93.0	yes	1.0
22	CSE	6.20821104006E11	M	42.0	41.0	47.0	98.0	yes	1.0
23	CSE	6.20821104007E11	M	45.0	46.0	49.0	100.0	yes	2.0
24	CSE	6.20821104008E11	M	48.0	46.0	48.0	90.0	no	0.0
25	CSE	6.20821104011E11	M	46.0	41.0	43.0	95.0	no	0.0
26	CSE	6.20821104012E11	M	41.0	43.0	45.0	98.0	no	0.0
27	IT	6.20821104071E11	M	47.0	46.0	48.0	99.0	no	0.0
28	IT	6.20821104072E11	M	45.0	43.0	43.0	90.0	no	0.0
29	IT	6.20821104073E11	M	45.0	46.0	48.0	98.0	yes	0.0
30	IT	6.20821104074E11	M	45.0	46.0	47.0	90.0	no	0.0

DATA VISUALIZATION OF DATA WAREHOUSE:





DATA VALIDATION:

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose **FilteredClassifier** -F "weka.filters.supervised.attribute.Discretize -R first-last -precision 6" -S 1 -W weka.classifiers.trees.J48 -- -C 0.25 -M 2

Test options

☐ Use training set

☐ Supplied test set **Set...**

☒ Cross-validation Folds: 10

☐ Percentage split %: 66

More options...

(Num) arrearcount

Start **Stop**

Result list (right-click for options)

- 15:03:08 - rules.M5Rules
- 15:03:21 - meta.FilteredClassifier**

Classifier output

```

=== Run information ===

Scheme:      weka.classifiers.meta.FilteredClassifier -F "weka.filters.supervised.attribute.Discretize -R first-last -precision 6" -S 1 -W weka.classifiers.trees.J48 -- -C 0.25 -M 2
Relation:    studentdetail
Instances:   51
Attributes:  9
  department
  Registernumber
  gender
  IAT1Mark
  IAT2Mark
  IAT3Mark
  Attendancepercentage
  Arrear
  arrearcount

Test mode:   10-fold cross-validation
  
```

Status: Problem evaluating classifier **Log**

DATA ANALYSE AND TESTING FOR DATA WAREHOUSE:

Weka Experiment Environment

Setup Run **Analyse**

Source

Got 51 results

File... Database... Experiment

Actions

Perform test Save output Open Explorer...

Configure test

Testing with: Paired T-Tester (corrected)

Select rows and cols: Rows Cols Swap

Comparison field: IAT1Mark

Significance: 0.05

Sorting (asc.) by: <default>

Test base: Select

Displayed Columns: Select

Show std. deviations: ☐

Output Format: Select

Result list

- 15:09:42 - Available resultsets
- 15:09:47 - Available resultsets
- 15:09:51 - Available resultsets
- 15:10:09 - department - 46
- 15:10:29 - department - 46
- 15:11:00 - IAT1Mark - Ranking
- 15:11:15 - IAT1Mark - Ranking
- 15:11:16 - IAT1Mark - Ranking
- 15:11:22 - IAT1Mark - Summary
- 15:11:45 - IAT1Mark - Summary
- 15:11:46 - IAT1Mark - Summary
- 15:11:52 - Available resultsets
- 15:11:58 - IAT1Mark - 46
- 15:11:59 - IAT1Mark - 46

Test output

Dataset (1) (2) (3) (4) (5) (6) (7) (8)

620821104002
620821104007
620821104008
620821104071
620821104073
620821104074
620821104076
620821104075
620821104072
620821104012
620821104011
620821104003
620821104077
620821104004
620821104006
620821104005
620821104078

Key:

- (1) 46
- (2) 47
- (3) 42
- (4) 41
- (5) 43
- (6) 44
- (7) 49
- (8) 45

RESULT:

Thus the data warehouse for real time application is explored, visualized, validated, analyzed and tested successfully.